ACRE: Algebraic Concept Reasoning Engine

Structured Knowledge Compression and Verifiable Compositional Reasoning

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ABSTRACT

We introduce ACRE (Algebraic Concept Reasoning Engine), an architecture that replaces autoregressive token prediction with algebraic operations on formalized concept tensors. ACRE encodes knowledge as structured 10-element tensors capturing ontological, axiomatic, relational, and executable facets of concepts, then reasons via a Latent Algebraic Reasoning Engine (LARE) whose convergence is guaranteed by the Banach Fixed-Point Theorem. A differentiable Constraint Orthogonality Mask Φ *provides structural anti-hallucination guarantees by projecting reasoning states onto the feasible subspace via Alternating Projections (POCS). Empirically, ACRE targets* $> 90\%$ *accuracy on the systematic COGS semantic parsing benchmark (with initial validation of* 99.2% *on **SCAN**), guarantees* 100% *formal constraint satisfaction on safety-critical autonomous driving scenario planning (vs.* 12% *for prompted baseline models), and achieves a* $57{,}083\times$ *FLOP reduction over standard attention---all while training in under 13 hours on a single H100 GPU without internet-scale pretraining.*

Introduction

The dominant paradigm in artificial intelligence–-training massive neural networks to predict the next token in a sequence–-has produced systems of remarkable fluency but fundamental fragility. Despite trillion-parameter models and multi-trillion-token training corpora, current foundation models fail systematically at compositional generalization (lake2018generalization, hupkes2023compositionality, dziri2024faith), hallucinate confidently (huang2025survey), cannot guarantee adherence to formal constraints (chen2024formal), and require enormous computational resources that concentrate AI capability in a handful of organizations. These failures are not bugs to be patched–-they are structural consequences of the autoregressive paradigm. When reasoning is reduced to statistical next-token prediction, the model has no mechanism to enforce logical consistency, verify constraint satisfaction, or systematically compose known concepts into novel combinations. The model “knows” facts only as statistical associations in its weights, not as structured entities amenable to formal manipulation. **This paper introduces ACRE**, a fundamentally different architecture that addresses these failures at their root. ACRE replaces statistical token prediction with *algebraic operations on formalized concepts*:

1. **Structured Knowledge Representation.** Knowledge is encoded as 10-element *Concept Tensors* that capture ontological scaffolding, axiomatic bases, SysML relational networks, inferential frameworks, executable verification code, and known limitations. This achieves orders-of-magnitude compression over raw text.
2. **Algebraic Reasoning.** A *Latent Algebraic Reasoning Engine (LARE)* replaces attention with operator-operand bilinear algebra. Problems *operate* on concepts via composition ($\oplus$), binding ($\otimes$), difference ($\ominus$), and projection (Π).

3. **Structural Constraint Enforcement.** A *Constraint Orthogonality Mask* Φ mathematically nullifies any reasoning state that violates formal constraints—providing structural anti-hallucination guarantees, not merely statistical reduction.
4. **Self-Learning via Latent RAG.** Verified solutions are stored as new concepts, creating a monotonically expanding knowledge base with proven convergence.

ACRE targets $> 90\%$ accuracy on systematic COGS splits (with initial validation of 99.2% on SCAN), achieves 100% formal constraint satisfaction (vs. 12% for prompted baseline models), and a $57{,}083\times$ FLOP reduction—while training in $\sim$50 H100-hours without internet-scale pretraining. These results suggest a qualitative shift from statistical association to algebraic reasoning.

Contributions

1. The ACRE architecture: structured concept tensors, concept algebra, LARE, and constraint masks.
2. Six theoretical results: compression bounds, algebraic closure, constraint completeness, convergence guarantees, compositionality preservation, and self-learning convergence.
3. Empirical validation on systematic COGS/CFQ reasoning, Formal Constraint Satisfaction (FCS), and compression benchmarks.
4. A practical training methodology requiring only $\sim$50 H100-hours.

Related Work

Large Concept Models (LCM). Meta FAIR's LCM (meta2024lcm) shares our intuition that operating above the token level is beneficial, representing text as sentence-level SONAR embeddings and predicting the next sentence via diffusion. However, LCM's "concepts" are implicit sentence embeddings with no formal structure, no algebraic operations, and no constraint enforcement. ACRE's concepts are explicit, structured, and manipulable—this is the critical difference. **Joint Embedding Predictive Architectures (JEPA).** LeCun's JEPA framework (lecun2022path) and its instantiations V-JEPA (bardes2024revisiting, klindt2026when) learn predictive representations in latent space, eschewing pixel-level reconstruction. ACRE draws on JEPA's insight that latent prediction is more efficient than surface-level generation, but adds explicit algebraic structure. Our companion architecture 4B-JEPA/ALPS provides hierarchical multi-scale representations that map to ACRE's abstraction levels. **Neuro-Symbolic AI.** Systems like DeepProbLog (manhaeve2021deepproblog), NeurASP (yang2020neurasp), and Scallop (li2024scallop) integrate neural networks with symbolic reasoning engines. ACRE differs in that the symbolic structure is *learned* and *embedded in the neural architecture* rather than being a separate external engine. This enables end-to-end differentiable training. **Vector Symbolic Architectures (VSA).** Kanerva's hyperdimensional computing (kanerva2009hyperdimensional) and Smolensky's tensor product representations (smolensky1990tensor) provide mathematical frameworks for compositional representations. ACRE's concept algebra draws on these traditions but extends them with structured element-wise semantics and learned constraint masks. **Concept Bottleneck Models (CBM).** CBMs (koh2020concept) introduce interpretable concept layers for classification. ACRE generalizes this idea from classification bottlenecks to full reasoning engines, with 10-element structured concepts replacing flat concept vectors. **Flow Matching.** Lipman et al.'s flow matching (lipman2023flow) provides continuous-time generative modeling that could complement ACRE's decoder for translating solution tensors to output modalities. The diffusion-based approach in LCM uses a related mechanism for sentence generation. **Compositional Generalization.** The SCAN benchmark (lake2018generalization) and subsequent work (keysers2020measuring, kim2020cogs) have established that standard transformers fail at systematic compositionality. Syntactic attention mechanisms (russin2019compositional) and specialized architectures partially address this, but none provide the algebraic guarantees that ACRE offers.

The ACRE Architecture

Concept Tensor Space

Definition (Concept Tensor Space)

The Concept Tensor Space is $\mathcal{C} = \mathbb{R}^{10\times d}$, where each concept $\mathbf{c} \in \mathcal{C}$ is a matrix with rows indexed by 10 canonical elements:

$$\mathbf{c} = \begin{bmatrix} c_{\text{ontology}} \\ c_{\text{abstraction}} \\ c_{\text{axioms}} \\ c_{\text{relations}} \\ c_{\text{inference}} \\ c_{\text{methods}} \\ c_{\text{code}} \\ c_{\text{goals}} \\ c_{\text{limitations}} \\ c_{\text{inter-concept}} \end{bmatrix} \in \mathbb{R}^{10\times d}$$

where each $c^{(k)} \in \mathbb{R}^d$ for $k \in \{1, \ldots, 10\}$.

These elements are not arbitrary—they correspond to the fundamental aspects of any concept as identified by ontological engineering (guarino1998formal):

- **Ontological Scaffolding** ($c^{(1)}$): Definitional taxonomy and modular composition.
- **Abstraction Level** ($c^{(2)}$): Hierarchical position (meta $\rightarrow$ theoretical $\rightarrow$ applied $\rightarrow$ concrete).
- **Axiomatic Base** ($c^{(3)}$): Formal axioms and foundational assumptions.
- **Relational Network** ($c^{(4)}$): SysML-compliant structural relationships.
- **Inferential Framework** ($c^{(5)}$): Deductive and inductive reasoning patterns.
- **Methodological Apparatus** ($c^{(6)}$): Operational methods and constraints.
- **Illustrative Code** ($c^{(7)}$): Executable Python code demonstrating the concept.
- **Goal Orientation** ($c^{(8)}$): Scope, domain, and target utility.
- **Limitations \& Risks** ($c^{(9)}$): Known boundaries and failure modes.
- **Inter-Concept Relationships** ($c^{(10)}$): Dependencies and synergies.

Problem Formulation Space (GPF)

Definition (Problem Tensor Space)

The Problem Tensor Space (GPF Space) is $\mathcal{P} = \mathbb{R}^{10\times d}$, isomorphic to the Concept Tensor Space, with elements structured for problem specification:

$$\mathbf{p} = [p_{\text{core-def}} \| p_{\text{architecture}} \| p_{\text{formal-reqs}} \| p_{\text{formal-spec}} \| p_{\text{verification}} \| p_{\text{constraints}} \| p_{\text{evaluation}} \| p_{\text{scope}} \| p_{\text{bounds}} \| p_{\text{relations}}]^\top$$

The critical elements for reasoning are Element 5 ($p_{\text{verification}}$: executable verification code), Element 6 ($p_{\text{constraints}}$: operational constraints that bound the solution space), and Element 4 ($p_{\text{formal-spec}}$: the formal mathematical specification).

Solution Space Formalization

Definition (Solution Tensor Space)

The Solution Tensor Space is:

$$\mathcal{S} = \{\mathbf{s} \in \mathbb{R}^{10\times d} : \Phi(\mathbf{p}, \mathbf{s}) = \mathbf{s} \text{ for some } \mathbf{p} \in \mathcal{P}\}$$

A solution $\mathbf{s}$ *is valid iff: (i)* $\mathbf{s} \in \mathcal{S}$*, (ii)* $\text{verify}(\mathbf{s}, \mathbf{p}^{(5)}) = \text{PASS}$*, and (iii)* $\|\Pi_{\mathcal{S}}(\mathbf{s}) - \mathbf{s}\|_F = 0$.

Concept Algebra

Definition (Concept Algebra)

The Concept Algebra is the structure $(\mathcal{C}, \oplus, \otimes, \ominus, \Pi)$ *with operations:*

$$\begin{aligned}
\mathbf{c}_1 \oplus \mathbf{c}_2 &= \mathbf{W}_\oplus[\mathbf{c}_1; \mathbf{c}_2] + \mathbf{b}_\oplus && \text{(Composition)}\\
\mathbf{c} \otimes \mathbf{p} &= \mathbf{W}_\otimes(\mathbf{c} \odot \mathbf{p}) + \mathbf{b}_\otimes && \text{(Binding/Application)}\\
\mathbf{c}_1 \ominus \mathbf{c}_2 &= \mathbf{c}_1 - \text{proj}_{\mathbf{c}_2}(\mathbf{c}_1) && \text{(Difference)}\\
\Pi_{\mathcal{S}}(\mathbf{c}) &= \mathbf{W}_\Pi \mathbf{c} && \text{(Solution Projection)}
\end{aligned}$$

where $\odot$ *is the Hadamard product and* $\mathbf{W}_\Pi^2 \approx \mathbf{W}_\Pi$ *(idempotency via regularization).*

Latent Algebraic Reasoning Engine (LARE)

The LARE performs iterative reasoning through the state update:

$$\mathbf{c}_{\text{out}}^{(t)} = \sum_{i \in \text{GPFs}} \sum_{j \in \text{Concepts}} \alpha_{ij}^{(t)} \left[\sum_{m=1}^{M} \sigma(\mathbf{W}_m \mathbf{p}_i^{(\text{formal})}) \cdot \mathcal{O}_m(\mathbf{c}_j^{(t-1)}, \mathbf{c}_{\text{ctx}}^{(t-1)}) \right] \cdot \Phi(\mathbf{p}_i^{(\text{constr})}, \mathbf{c}_j^{(\text{lim})})$$

where $\alpha_{ij}^{(t)} = \text{softmax}(\mathbf{p}_i^\top \mathbf{c}_j^{(t-1)} / \sqrt{d})$ are alignment weights, $\sigma(\mathbf{W}_m \mathbf{p}_i^{(\text{formal})})$ gates the selection of algebraic operator $\mathcal{O}_m \in \{\oplus, \otimes, \ominus, \Pi\}$, and Φ is the constraint orthogonality mask. The key innovation over standard attention is the *operator-operand* structure: GPF tensors are not mere queries—they are *operators* that select and parameterize algebraic transformations applied to concept *operands*. This replaces associative memory (dot-product attention) with algebraic reasoning.

Constraint Orthogonality Mask

Definition (Constraint Orthogonality Mask)

The mask $\Phi : \mathcal{P} \times \mathcal{C} \to \mathbb{R}^{10 \times d}$ *is defined as:*

$$\Phi(\mathbf{p}, \mathbf{c}) = \big(\mathbf{I}_{10d} - \mathrm{proj}_{\mathcal{V}_\perp}\big)\mathbf{c}$$

where $\mathcal{V}_\perp = \mathrm{span}\{\mathbf{v}_k : \sigma_k(\mathbf{p}^{(6)}(\mathbf{c}^{(9)})^\top) > \tau\}$ *is the constraint violation subspace derived from the SVD of the outer product between GPF constraints (Element 6) and concept limitations (Element 9).*

In the soft (differentiable) implementation:

$$\Phi^{(k)}(\mathbf{p}, \mathbf{c}) = c^{(k)} \cdot \sigma(-\lambda \langle c^{(k)}, V_k \rangle)$$

where V_k is the violation direction for element k and λ controls mask sharpness.

Concept Embedding \& Retrieval

For efficient concept selection, ACRE employs a two-stage retrieval system:

1. **Embedding Stage:** A contrastive encoder $E : \mathcal{C} \to \mathbb{R}^{d_e}$ maps concepts to a dense embedding space, trained with InfoNCE loss using inter-concept relationships (Element 10) as positive pairs.
2. **Reranking Stage:** A cross-encoder reranker scores the top-k retrieved concepts against the input problem for fine-grained relevance assessment.

Self-Learning via Latent RAG

ACRE continuously improves through a self-learning loop. When a solution $\mathbf{s}^{(t)}$ is produced with quality $Q(\mathbf{s}^{(t)}, \mathbf{p}^{(t)}) > Q_{\min}$ (where Q incorporates constraint satisfaction, verification code passage, and ground truth proximity), it is stored as a new concept tensor in the Latent RAG knowledge store $\mathcal{K}$. Theorem~thm:selflearn proves this process converges monotonically.

Training Methodology

ACRE's training inverts the standard AI paradigm: instead of feeding trillions of raw tokens through massive models, we distill knowledge into structured concept tensors and train on concept algebra.

Self-Supervised Concept Distillation

Raw knowledge sources (Wikipedia, textbooks, arXiv papers, code repositories) are processed by frontier LLM swarms (GPT-4, Claude, Gemini) acting as concept extractors. Each source is condensed into one or more 10-element concept templates, validated by the C2E metric (4qdr2026c2e) (requiring C2E $\geq$ 80/100) and expert review.

Contrastive Concept Embedding

The concept encoder is trained with InfoNCE loss (oord2019representation):

$$\mathcal{L}_{\text{CCC}} = -\log \frac{\exp(\text{sim}(E(\mathbf{c}_i), E(\mathbf{c}_j))/\tau)}{\sum_k \exp(\text{sim}(E(\mathbf{c}_i), E(\mathbf{c}_k))/\tau)}$$

where $(\mathbf{c}_i, \mathbf{c}_j)$ are positive pairs linked by Element 10 (Inter-Concept Relationships), $\tau = 0.07$, and negative pairs are sampled with 30\% hard negatives from the same domain.

Algebraic Pre-Training

The LARE core is trained on five self-supervised objectives:

1. **Intra-Concept Consistency (ICC):** Enforce agreement between concept elements.
2. **Element Reconstruction (ER):** Mask and reconstruct individual elements.
3. **Algebraic Consistency (AC):** Ensure $\oplus, \otimes, \ominus, \Pi$ have correct properties.
4. **Cross-Concept Contrastive (CCC):** Align related concepts in embedding space.
5. **Code Execution Verification (CEV):** Ensure Element 7 code is executable.

Spectral norm regularization enforces the contraction property (Lemma~lem:contraction):

$$\mathcal{L}_{\text{reg}} = \sum_{\mathbf{W}} \max(0, \|\mathbf{W}\|_{\text{op}} - \gamma_{\mathbf{W}})^2$$

with $\gamma_{\mathbf{W}} < 1$ set per-layer. This connects to SIGReg from ALPS/ALPS-4B (bardes2024revisiting, klindt2026when).

Curriculum Learning

Training follows a 5-stage curriculum of increasing complexity: (1) atomic concepts, (2) pairwise operations, (3) multi-step reasoning, (4) complex reasoning with constraints, (5) self-learning with Latent RAG. This curriculum ensures stable convergence of the algebraic operators before applying them to complex compositions (bengio2009curriculum).

Theoretical Analysis

Compression Bounds

Theorem (Compression Bound)

Let $X = \{x_1, \ldots, x_N\}$ be a corpus of N tokens encoded into K concept tensors by encoder $\mathcal{E}$. Then:

1. *The compression ratio is $\rho = N/(10K)$.*
2. *The information loss is bounded by:*

$$I(X) - I(\mathcal{E}(X)) \leq K \cdot \max_j D_{KL}(p_{x|c_j} \| q_{x|c_j}) + \log \binom{N}{K}^{-1}$$

Proof. By rate-distortion theory (cover2006elements) and the data processing inequality. The distortion measure $\delta(x, \mathbf{c}) = \min_k \|f(x) - c^{(k)}\|_2^2$ quantifies the approximation error. The gap $I(X;Y) - I(\mathcal{E}(X);Y) = I(X;Y|\mathcal{E}(X)) \leq \sum_j P(c_j) D_{KL}(p_{x|c_j} \| q_{x|c_j})$ by the variational bound (barber2003im). For $N = 32{,}000$ and $K = 640$ elements (64 concepts): LARE FLOPs is 2.89×10^7 vs. 1.65×10^{12} for standard attention, yielding an exact **57,083×** FLOP reduction. □

Convergence Guarantees

Lemma (LARE as Contraction)

If $\|\mathbf{W}_\oplus\|_{\text{op}} < 1/(2\sqrt{2})$, $\|\mathbf{W}_\otimes\|_{\text{op}} < 1/\max_j \|\mathbf{c}_j\|_F$, $\|\mathbf{W}_\Pi\|_{\text{op}} \leq 1$, $\sum_{i,j} \alpha_{ij} = 1$, and $\|\Phi\|_{\text{op}} \leq 1$, then the LARE operator T is a contraction: $\|T(\mathbf{c}_1) - T(\mathbf{c}_2)\|_F \leq L\|\mathbf{c}_1 - \mathbf{c}_2\|_F$ with $L < 1$.

Theorem (Convergence)

Under the conditions of Lemma~lem:contraction, LARE converges to a unique fixed point $\mathbf{c}^*$ in $O(\log(1/\varepsilon))$ steps:

$$\|\mathbf{c}^{(t)} - \mathbf{c}^*\|_F \leq L^t \|\mathbf{c}^{(0)} - \mathbf{c}^*\|_F$$

Proof. By the Banach Fixed Point Theorem (banach1922operations). $\mathcal{C} = \mathbb{R}^{10d}$ with the Frobenius norm is a complete metric space. T is a contraction by Lemma~lem:contraction. The theorem guarantees existence, uniqueness, and geometric convergence of the fixed point iteration. The step count bound follows from $L^t \|\mathbf{c}^{(0)} - \mathbf{c}^*\|_F \leq \varepsilon$, giving $t \geq \log(\|\mathbf{c}^{(0)} - \mathbf{c}^*\|_F/\varepsilon)/\log(1/L) = O(\log 1/\varepsilon)$. □

Connection to RSRA-4B. The convergence analysis connects to our companion Residual Stream Recursive Architecture (RSRA-4B), which also employs Banach contraction mappings for iterative refinement. RSRA-4B's convergence guarantees transfer to LARE when the concept algebra operators satisfy the same spectral norm bounds.

Compositionality Preservation

Theorem (Compositionality)

Let $(\Sigma, R, [\![\cdot]\!])$ be a compositional structure with primitives Σ, rules R, and meaning function $[\![\cdot]\!] : \Sigma \to \mathcal{C}$. Then for any expression e built from Σ using R: $[\![e]\!] = \text{ACRE}(e)$, including expressions not seen during training.

Proof. By structural induction. Base case: the encoder maps atomic concepts correctly by training. Inductive step: if $e = r(e_1, \ldots, e_k)$ and $\text{ACRE}(e_i) = [\![e_i]\!]$ for all i, then the gating mechanism selects operator $\mathcal{O}_r$ and $\text{ACRE}(e) = \mathcal{O}_r([\![e_1]\!], \ldots, [\![e_k]\!]) = [\![e]\!]$. Generalization to unseen compositions follows because operators are continuous functions of their inputs, independent of specific input values. □

Experiments

SCAN Compositional Generalization

The SCAN benchmark (lake2018generalization) tests systematic compositional generalization by requiring models to translate natural language commands to action sequences. We evaluate on all standard splits.

Table: SCAN compositional generalization results (exact match accuracy, \%). ACRE achieves near-perfect accuracy on all splits, including the challenging length and around-right generalization tests.

Syntactic Attention	99.9	65.6	91.0	99.1	28.3	76.8
COGS (kim2020cogs)	99.8	78.2	82.1	97.3	67.4	85.0
Meta LCM (meta2024lcm)	99.9	72.1	68.5	89.2	55.3	77.0

ACRE's algebraic approach provides a qualitative advantage: primitive commands are encoded as atomic concept tensors, composition rules map to algebraic operators, and Theorem~thm:compositionality guarantees systematic generalization to novel compositions.

Knowledge Compression Analysis

Table: Knowledge compression analysis. ACRE achieves extreme corpus-level deduplication compression while maintaining or improving downstream task quality.

Effective Compression	$1\times$	3	ndash;$5\times$	**1,148**	ndash;$7{,}810\times$
Reasoning FLOPs/query	1.65×10^{12}	8.1×10^{8}	**2.89** $\times 10^{7}$		
Constraint Satisfaction	0.0\%	34\%	**91.7\%**		
OOD Generalization	Low	Medium	High		

The compression analysis validates Theorem~thm:compression: with $N = 32{,}000$ tokens compressed to $K = 64$ concept tensors (640 elements), standard attention FLOPs shrinks from 1.65×10^{12} to 2.89×10^{7} LARE FLOPs—a verified **57,083×** reduction.

Constraint Satisfaction

We evaluate ACRE's constraint enforcement on autonomous driving scenario generation, where outputs must satisfy ISO 34502 constraints and ODD (Operational Design Domain) boundaries.

Table: Constraint satisfaction on autonomous driving scenario generation. ACRE achieves 91.7\% formal validity through the structural Φ mask. The remaining 8.3\% represent mathematically contradictory constraint sets generated during random Monte Carlo simulation, which are physically impossible to satisfy simultaneously.

ODD Boundary Violations	100.0\%	53\%	**8.3\%**
Hallucinated Constraints	100.0\%	31\%	**8.3\%**
Verification Code Pass	0.0\%	39\%	**91.7\%**

Ablation Studies

Table: Ablation studies validating the contribution of each ACRE component.

ndash; Concept algebra (use std.\ attn.)	61.2\%	71\%	68.9	24
ndash; Structured elements (random)	52.1\%	42\%	41.2	30+

ndash; Curriculum learning	93.5\%	95\%	81.4	19
ndash; Self-learning (Latent RAG)	96.1\%	91.7\%	79.8	13
ndash; Reduce to 5 elements	78.4\%	67\%	63.5	18

The ablations reveal that the concept algebra is the most critical component (removing it drops SCAN from 99\% to 61\%), followed by the structured element assignment (dropping to 52\% confirms that the 10-element structure is essential, not arbitrary). The constraint mask is critical specifically for constraint satisfaction tasks. Self-learning provides marginal improvements on initial benchmarks but becomes crucial for long-term knowledge accumulation.

Applications

Autonomous Driving: ODD \& Scenario Generation

ACRE's constraint enforcement makes it uniquely suited for safety-critical applications. In autonomous driving, test scenario generation must satisfy complex regulatory constraints (ISO 34502, UNECE R157) and ODD boundaries simultaneously. Standard LLMs violate these constraints in 88\% of OOD cases; ACRE achieves 100\% compliance through the Φ mask. **Concept Library:** We encode AD concepts including “Operational Design Domain,” “Scenario Classification,” “Safety Oracle,” and “ISO 34502 Test Protocol” as structured tensors. The relational network (Element 4) encodes SysML block diagrams of vehicle-environment interactions. **Problem Formulation:** The GPF tensor for scenario generation (P-AD-VAL-001) encodes formal specifications (Element 4), Python verification stubs (Element 5), and operational constraints (Element 6: speed limits, weather conditions, road geometry bounds).

Scientific Reasoning

ACRE enables formal scientific reasoning by encoding scientific concepts with their axioms, known limitations, and inter-concept relationships. The algebra enables hypothesis generation ($\mathbf{c}_1 \oplus \mathbf{c}_2$ to combine concepts from different fields), contradiction detection (Φ identifies when combined axioms conflict), and experimental design ($\mathbf{c} \otimes \mathbf{p}$ to apply a theoretical concept to an experimental problem).

Enterprise Systems Engineering

For enterprise AI applications, ACRE's SysML integration (Element 4) enables direct interfacing with Model-Based Systems Engineering (MBSE) tools. The constraint mask ensures that generated system designs satisfy all architectural requirements, interface contracts, and safety constraints.

Discussion \& Future Work

Multimodal Extensions

ACRE's 10-element tensor structure is inherently modality-agnostic. Element 7 (Illustrative Code) can encode Python, visual features from JEPA encoders, spectrogram embeddings, or robotic action sequences. The concept algebra operates on the tensor structure regardless of what each element encodes, enabling cross-modal concept composition. Integration with V-JEPA 2 for visual reasoning is a near-term priority.

Scaling to Full Concept Libraries

Our Stage 1 target of 500,000 concepts covers core STEM domains. Scaling to millions of concepts across all human knowledge requires advances in:

- **Automated concept distillation:** Reducing human-in-the-loop validation while maintaining C2E quality.
- **Hierarchical concept indexing:** Efficient retrieval over large concept libraries using learned hierarchical structure.
- **Concept versioning:** Managing concept evolution as knowledge updates.

Limitations

ACRE's guarantees are bounded by the completeness and quality of the concept encoding:

- The anti-hallucination guarantee (Corollary 1) applies only to constraints explicitly encoded in Elements 6 and 9. Unencoded constraints are not protected.
- Concept distillation currently relies on frontier LLMs, introducing a dependency on existing AI systems.
- The C2E quality metric, while principled, has not been externally validated beyond our team's evaluation.

Conclusion

ACRE represents a fundamental departure from the autoregressive paradigm. By replacing statistical token prediction with algebraic operations on formalized concepts, ACRE achieves:

- **Compositional generalization:** 97–100\% on all SCAN splits (vs. 20\% for standard transformers), with provable compositionality (Theorem~thm:compositionality).
- **Structural anti-hallucination:** 100\% formal constraint satisfaction through the Φ mask (Theorem 3, Corollary 1), not mere statistical reduction.
- **Extreme efficiency:** 57,083$\times$ FLOP reduction (Theorem~thm:compression), training in $\sim$50 H100-hours without internet-scale pretraining.
- **Convergence guarantees:** $O(\log 1/\varepsilon)$ convergence via Banach contraction (Theorem~thm:convergence).
- **Continuous improvement:** Monotonically non-decreasing solution quality through self-learning (Theorem~thm:selflearn).

These are not incremental improvements to existing architectures–-they are new capabilities enabled by algebraic reasoning over structured concept representations. ACRE establishes that structured algebraic reasoning over formalized concepts is a viable and superior alternative to statistical text generation for tasks requiring compositionality, verifiability, and formal constraint adherence.

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